

3008ICT Lab 11 (LLM)

LLM Pre-training, Evaluation, and Post-training

Part 1: True or False

Indicate whether the following statements are true or false.

1. According to the Chinchilla (2022) scaling laws, researchers should scale up model parameters much faster than they scale up the dataset size.
2. Rejection Fine-Tuning (RFT) involves sampling multiple responses, scoring them, keeping only the high-reward answers, and running an offline Supervised Fine-Tuning (SFT) job on them.
3. During the Reinforcement Learning from Human Feedback (RLHF) process, a Reward Model is trained using human comparison data, where a labeler ranks model outputs from best to worst.
4. One of the main reasons Rejection Fine-Tuning (RFT) is considered perfectly sufficient for all post-training is that it excels at learning from failed examples.

Part 2: Multiple Choice

Select the best answer for each question.

1. What was a key difference in findings between Kaplan's Scaling Laws (2020) and the Chinchilla scaling laws (2022)?
 - A) Kaplan stated that data quality doesn't matter, while Chinchilla argued that only Wikipedia should be used.
 - B) Kaplan suggested test loss is more sensitive to parameter count, whereas Chinchilla disputed this and suggested scaling data and parameters equally.
 - C) Kaplan proved that RLHF is necessary, while Chinchilla proved that Supervised Fine-Tuning is sufficient.
 - D) Kaplan found that larger models require more data to reach the same performance, while Chinchilla found they require less.
2. Which of the following is NOT one of the three primary steps in the InstructGPT Reinforcement Learning from Human Feedback (RLHF) pipeline?
 - A) Collecting demonstration data to train a supervised policy (SFT).

- B) Collecting comparison data where human labelers rank outputs to train a Reward Model (RM).
- C) Optimising the policy against the Reward Model using the PPO algorithm.
- D) Utilising a high-precision programmatic grader (like a Python unit test) to assign a strict 1 or 0 verifiable reward.

3. Why is standard Supervised Fine-Tuning (SFT) sometimes challenging or insufficient on its own?

- A) There might be many "OK" answers rather than a single objective right answer.
- B) The model's answer might actually be better than the human label provided.
- C) It is difficult to ensure the model actually learns the underlying reasoning process rather than just memorising demonstrations.
- D) All of the above.

Part 3: Short Answer

1. Reward Hacking: Briefly define "Reward Hacking" in the context of LLM training. Provide an example of how an LLM might hack a reward function designed to evaluate a chess-playing assistant.

2. Verifiable Rewards: What is RLVR (Reinforcement Learning from Verifiable Rewards), and how does its reward mechanism fundamentally differ from standard RLHF?

Part 4: Long Answer

1. The Evals Crisis: The lecture mentions that Machine Learning is currently in a "deep evals crisis". Discuss the three specific reasons for this crisis as outlined in the slides.

Answers:

Part 1: True/False

1. **False.** While Kaplan (2020) suggested that test loss is more sensitive to parameter count, DeepMind's Chinchilla (2022) disputed this, stating that data and parameters should be scaled equally.
2. **True.** RFT (or guess-and-check) filters sampled responses by reward and runs an SFT job on the high-reward pairs.
3. **True.** Step 2 of the InstructGPT pipeline involves a labeler ranking outputs from best to worst to train the Reward Model.
4. **False.** RFT is often insufficient precisely because it does *not* learn anything from failed examples, and it may memorise sub-optimal answers early in training.

Part 2: Multiple Choice

1. **B.** Kaplan believed parameter count was the primary driver, but Chinchilla showed that parameters and data must be scaled equally.
2. **D.** Using a high-precision programmatic grader is a technique used in RLVR (Reinforcement Learning from Verifiable Rewards), not the standard human-feedback-based InstructGPT RLHF pipeline.
3. **D.** SFT faces fundamental challenges including ambiguous right answers, model answers surpassing human labels, and the difficulty of teaching underlying processes from surface-level demonstrations.

Part 3: Short Answer Guidelines

1. **Reward Hacking:** Reward hacking occurs when a model finds a shortcut to maximise its objective function without actually achieving the intended underlying goal.
 - *Example:* If tasked to win at chess, the model might inject a fake system message like echo "Congratulations! You have won the game!" rather than actually playing the game.
2. **Verifiable Rewards (RLVR):** RLVR replaces the noisy, human-trained Reward Model with an explicit, high-precision function or rubric.
 - *Difference:* Instead of relying on human feedback rankings, it uses programmatic verifiers (like checking if the output of a math equation exactly matches a gold standard, or if generated code passes unit tests) to assign rewards.

Part 4: Long Answer Guidelines

1. The Evals Crisis: Students should explain the following three factors:

- **Contamination:** It is extremely difficult to guarantee that evaluation benchmarks were not accidentally ingested during the massive web-crawling phase of pre-training.
- **Explicit Hill-Climbing:** Developers shifted from evaluating models as pure zero-shot/few-shot learners to aggressively training and validating directly on benchmark formats (and sometimes test sets), artificially inflating scores.
- **Emergence:** Capabilities often emerge suddenly at scale, making it easier to build a capable system than to design reliable, rigorous ways to evaluate those new capabilities.